

Subject Description Form

Subject Code	ABCT4115
Subject Title	Bioinformatics
Credit Value	3
Level	4
Pre-requisite	DNA Technology, Biochemistry, Molecular Biology, Cell Biology
Co-requisite	NIL
Objectives	The subject introduces students to basic principles of bioinformatics, paying particular attention to practical aspects. Students will be able to learn how to select and use proper software for bioinformatics analysis. In addition, the latest topics like biological AI models will also be covered.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand basic concepts and methods in bioinformatics, including working principles of DNA sequencers and biological mass spectrometry and its applications in proteomics and metabolomics, sequence assembly and alignment, gene expression quantification, gene regulatory elements, protein and metabolite identification and quantification, key elements of proteomics and metabolomics workflows. - Use appropriate databases, biological datasets, web servers, and software to investigate gene and protein functions. - Perform essential statistics in bioinformatics such as multivariate analysis, clustering, and dimensionality reduction. Brief understanding of programming in bioinformatics analysis. - Visualize results from bioinformatics data analysis and interpret the biological implications. Be aware of the developing trend in bioinformatics and the impact on science, humanity and society.
Subject Synopsis/ Indicative Syllabus	Principles of sequencing technologies, sequence assembly, sequence alignment, variant identification and annotation, DNA foundation models, genome editing Hands-on tutorial of FastQC, Megahit, BLAST Principles of RNA sequencing, RNA read mapping, gene expression quantification, differential gene expression analysis, gene set enrichment analysis Hands-on tutorial of DESeq2, g:Profiler, MetaScape, GSEA Basic concepts of gene regulatory elements, analysis of transcription factors, DNA methylation, chromatin accessibility, histone modification, three-dimensional genome Single cell sequencing technologies, characteristics of single cell data, dimensionality reduction, cell type annotation, single cell foundation models, spatial omics techniques Hands-on tutorial of single cell RNA sequencing data analysis using Seurat

	Basics of proteomics, basic theory of biological mass spectrometry (MS) and liquid chromatography-MS instruments, protein identification, quantitative proteomics, post-translational modifications (PTMs), data analysis and bioinformatics of MS data. Basics of metabolites and techniques in metabolomics, pre-process, process and analyse metabolomics data using univariate and multivariate data analysis, metabolomics downstream analyses for metabolic pathway and network analysis, and brief introduction of multi-omics. Hands-on tutorial of metabolomics data analysis using MetaboAnalyst																																																				
Teaching/Learning Methodology	Lectures will be used as the major content-delivery tool. Students will be able to learn relevant software and programming by practice in the tutorial sessions and assignments. A self-learning component will be included to write a term paper on an interesting application of bioinformatics. Reference materials will be distributed before and in class, from library and from Internet where software is constantly updated. Computational lab practicals will be used to learn how to perform database searching and gene/protein function analysis.																																																				
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th><th></th></tr> </thead> <tbody> <tr> <td>1. Attendance</td><td>5%</td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>2. Quizzes</td><td>10%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr> <tr> <td>3. Term report</td><td>25%</td><td></td><td></td><td>✓</td><td>✓</td><td></td></tr> <tr> <td>4. Practical</td><td>60%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr> <tr> <td>Total</td><td>100 %</td><td colspan="5"></td></tr> </tbody> </table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Quizzes are used to gauge how much students have learned in the basic concepts of bioinformatics, genomics, and proteomics. Writing skills will be assessed in all the assessment tasks and methods. Term report is used to assess whether students can identify recent developments in bioinformatics. The practicals and lab reports demand students to demonstrate their competence in using software and in the interpretation and analysis of data. Students are required to attend at least 75% of scheduled sessions for the subject. Students fail to fulfill the attendance requirement will lose the 5% attendance score.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d		1. Attendance	5%						2. Quizzes	10%	✓	✓	✓	✓		3. Term report	25%			✓	✓		4. Practical	60%	✓	✓	✓	✓		Total	100 %					
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Student Study Effort Expected	Class contact:																																																				
	Lectures					24 Hrs.																																															
	Tutorials					6 Hrs.																																															
	Practical (computer applications)					12 Hrs.																																															

	Other student study effort:	
	Self-study	60 Hrs.
	Total student study effort	102 Hrs.
Reading List and References	Supplementary 1. Baxevanis, A. D., Bader, G. D., & Wishart, D. S. (Eds.). (2020). <i>Bioinformatics</i> (Fourth edition.). Wiley. 2. Sanjeeva Srivastava. From Proteins to Proteomics: Basic Concepts, Techniques, and Applications. CRC Press; 2022 3. Jerzy Adamski. Metabolomics for Biomedical Research. London: Academic Press; 2020 4. www.ncbi.nlm.nih.gov 5. https://bis.zju.edu.cn/bioinfo101/ (in Chinese)	